



Analysis of the Health Network of Metropolitan Lima Against Large-Scale Earthquakes

Miguel Nunez-del-Prado^(✉)  and John Barrera

Universidad del Pacífico, Lima, Peru
m.nunezdelpradoc@up.edu.pe

Abstract. Peru is a highly seismic country located in the Ring of Fire, making it vulnerable to earthquakes and tsunamis. In the present work, we examined Lima's health system capacity from three different and complementary points of view. We first analyze the Hospital Treatment Capacity (HTC) of 41 hospitals of II and III categories from EsSalud and MINSA in Lima, Peru. Second, we computed the hospitals' coverage area and their citizens' health demand as the aftermath of an earthquake of 8 Mw magnitude. Finally, an accessibility simulation to reach the hospitals was performed, taking into account real traffic jams conditions and street degradation. This document aims to provide elements for the strengthening of the Peruvian fragile health system.

Keywords: Health network · Earthquakes · Graph theory · Optimization

1 Introduction

Peru is a highly seismic country, located in the so-called fire belt between the Nazca plates and the South American plate, where the Geophysical Institute of Peru (*i.e.* IGP) identified 33 different sources of earthquakes [7]. Mainly, Lima is within the source number 24 of earthquakes, namely *Region 24*. Besides, the study of Pulido *et al.* confirms this source of earthquakes with observations suggesting an aftershock of the Lima earthquake of 1746 [18].

The Ministry of Housing, Construction, and Sanitation of Peru consider that the m^2 of housing construction is valued at \$580 US Dollars [23] in Lima. Based on the coefficients determined in [7], the cost of building a square meter of a hospital in the Lima region would be around \$870 US Dollars for the 2019–2020 fiscal period. Additionally, literature studies agree that the Peruvian hospital system is fragile due to its age and lack of investment in decades [3, 10, 13, 19]. To this, we add the allocation of hospital demand by political divisions (*i.e.*, districts) that unrealistically capture hospitals' demand. Moreover, in a context like Peru, where we are close to the queue in terms of public health investment in Latin America, the health system is not prepared to receive a large number of

expected cases due to the magnitude of the earthquake. Therefore, it is relevant to analyze the state of the health system.

In this study, we propose to analyze hospitals' demand under the hypothesis that after an earthquake of 8 Mw of magnitude, only twenty hospitals will be operational. In a second stage, we will proceed to analyze the resistance of the access roads to the hospitals against the degradation they may suffer due to an earthquake of such magnitude. Thus, this evaluation will take into account the time of traffic that a vehicle may encounter when transiting through Lima on a Friday between noon and two o'clock in the afternoon. Finally, this document is organized as follows. Section 2 describes the works found in the literature. Section 3 details both the hospital demand and route degradation estimation, while Sect. 4 presents the results found. Finally, Sect. 5 concludes this study and presents new avenues of research to be explored.

2 Related Works

In this section, we describe the different studies on demand for hospitals after an earthquake and hospital capacity evaluation. For example, Bambarén *et al.* [3] proposes a probabilistic model to determine the demand for hospital care after a significant earthquake in metropolitan Lima. The authors consider the demand for hospital care, time of arrival at the hospitals, type of medical treatment; reason for hospital admission; and the need for specialized care such as hemodialysis, blood transfusions, and surgical procedures. The authors estimated that between 23 328 and 178 387 wounded would go to hospitals, of which between 4 666 y 121 303 would require hospital care, while between 18 662 and 57 084 could be treated as outpatients. It was estimated that there would be an average of 8 768 cases of crush syndrome and 54 217 cases of other health problems. Enough blood would be needed to 8 761 injuries in the first 24h. Also, it was expected that there would be a shortage of hospital beds and operating theatres due to high demand. Unfortunately, the authors do not detail the mathematical model used to obtain these values.

Liguori *et al.* [13], propose a model to estimate the capacity of a hospital in metropolitan Lima after a seismic event. To achieve this objective, they evaluated hospitals' seismic response capacity considering variables such as seismic risk, hospital integrity, supplies, hospital treatment capacity, and resources for emergency response coordination. This study determined that structural damage due to the hospital's age conditions the hospital's proper functioning in the event of an earthquake. Also, the priority hospitals would not be able to handle the estimated between 4 666 and 121 303.

In the evaluation of hospitals raised in [14], a five-step qualitative methodology is presented. The first step characterizes the Health Network in provinces. In the second step, the different phenomena that could occur in the evaluated territory are evaluated. In the third step, they determine the magnitudes or severities of the different phenomena evaluated in the previous step. In the fourth step, the vulnerability of that health network to the phenomena and levels determined

is analyzed. Finally, the level of risk is determined in a double-entry matrix to characterize it in terms of danger and vulnerability, depending on low, medium, and very high. However, one of the limitations of this evaluation is the lack of a quantitative model. According to the Hospital Safety Index guide (*ISH*) [16], this measure helps to quantify how robust a hospital is in the face of an adverse situation such as a natural disaster.

Liguori *et al.* [11] do a study on the losses, safety and functionality of 41 hospitals in metropolitan Lima using the *Comprehensive Approach for Probabilistic Risk Assessment* (CAPRA) [5]. The CAPRA methodology evaluates physical and human economic losses by analyzing hazards, exposure, and vulnerability. This work showed that for earthquakes with a return period of 500 and 1000 years, only 14% and 18% of the analyzed infrastructure would be operational, respectively. Based on this work, Liguori *et al.* [10] proposes a quantitative methodology to assess the care capacity of a hospital *HTC* in an earthquake scenario. The adapted model of [12] contemplates for the Peruvian case an efficiency factor α , human component β , number of operating environments in normal conditions γ , the average time for a surgery t , the operational probability of the physical component δ , and operational autonomy k .

Santa-Cruz *et al.* [17,20] presents the seismic risk assessment of hospitals in the city of Lima. This work aims to evaluate the seismic risk of Lima's health system on a metropolitan scale. The seismic risk analysis was carried out using a comprehensive approach for the Probabilistic Risk Assessment Methodology (CAPRA). A total of 41 hospitals were evaluated to obtain risk indicators in terms of economic losses. The results show that after an earthquake of magnitude 8.2 Mw with an epicenter on the coast of Lima, 85% of the hospital buildings have more than 10% of structural damage. This study seeks to understand and evaluate hospitals' seismic risk at the municipal level to define measures, priorities, and actions for the future development of Lima's health system.

Mulyasari *et al.* [15] assesses how well hospitals in the Tohoku and Nankai regions are prepared for earthquakes. The authors use a questionnaire survey consisting of six parameters and 21 indicators of the "four pillars of hospital preparedness," including structural, non-structural, functional, and human resources. The results show that most hospitals surveyed comply with functional preparedness, which is useful during the emergency period of a disaster, while the other three pillars (structural, non-structural, and human resources) need to be strengthened. This study helps assess the state of disaster preparedness and the gaps for these hospitals, drawing lessons from the Great Earthquake and Tsunami in eastern Japan's Tohoku area. The gaps found are used as a starting point to indicate how to improve preparedness and resilience to future disaster risks.

The work of Carrasco *et al.* [6] computes travel times for different cities in Peru to primary, secondary, and tertiary health service. Authors use satellite images to build a 500×500 grid to compute the time to travel in each grid. For computing travel time, authors rely on Google Earth Engine finding travel times ranging from 39, 152, and 448 min.

The following section details the proposed methodology for determining hospital capacity, demand for care, and post-seismic care facility accessibility assessment.

3 Methodology

In this section, we detail the methodology to be followed, which consists of three phases.

1) *Scenario Generation.*- For the scenario generation, we use the data collected in the literature [11,14,17,20] to calculate the HTC of the 41 MINSA and EsSalud for both category II and category III public hospitals considered in this study. Taking the model proposed in [11], which we extended to include the number of wards the hospital has δ and operating time k autonomous after an earthquake (*c.f.*, Eq. 1).

$$HTC(\delta, k) = \alpha\beta\frac{\gamma^\delta}{t}k \quad (1)$$

Where the emergency plan efficiency α , and the preparedness of health care professionals, β , are set at 0.8 in a range of 0–1. With regard to operational autonomy, different scenarios were considered in hours $k = \{12, 24, 48, 72, 120\}$, while γ was determined using the performance point assessment [1,8] to obtain an estimate of 23%. In our case, we will take different scenarios considering the number of buildings δ essential E , and some essential buildings plus operational wards ER (*i.e.*, $E \geq ER$), to determine the number of people who can attend, with different times of autonomy. As for the average duration of surgical intervention. As for the average duration of a surgical intervention t , they decided two hours on average based on Lupoi's study *et al.* [12]. In the case of other types of care that are not surgeries (*i.e.* operaciones), it was estimated that the average time is $t = 1.5h$. This time was found based on telephone interviews with doctors working at the Rebagliati Hospital, the Collique Hospital, and the Cusco Regional Hospital in Peru. Finally, based on the different scenarios, we generated different hospital care capacity (HTC) rankings to obtain the top 20 categories II and III hospitals that will attend in that scenario.

2) *Estimating Demand for Medical Care.*- Regarding the demand for hospital services after an earthquake of grade 8 Mw, we collected information from the literature summarized in Table 1. This table shows the minimum, maximum and average for different types of hospital services, such as: *surgeries*, *medical care*, *outpatient care*, *crush syndrome* and *others* types of hospital care. Traditionally, demand estimation is performed by districts in a static way [3]. However, the before-mentioned assumption is unrealistic. Thus, after an earthquake, affected individuals try to reach the nearest hospital, whether it is in the same district of residence.

Table 1. Summary of estimated demand for the number of surgical procedures (*surgeries*), (*medical care*), (*outpatient care*), or (*crush syndrome*) and other types of care *others* after an 8 Mw earthquake.

Demand	Min	Average	Max
Surgeries [10]	2 481	33 499	64 515
Medical care [3]	4 666	62 985	121 303
Outpatient care [3]	18 662	37 873	57 084
Crush syndrome [3]			8 768
Others [3]			54 217
Total			305 887

To determine the demand, we took as input the top 20 hospitals' locations with the highest HTC to determine their coverage area using Voronoi diagrams [2]. Thus a coverage area can comprise fractions of different districts. Finally, to estimate the demand by km^2 , we only consider the areas of the districts delimited by the streets. For this we built the Lima street network, extracted from the Open Street Maps API¹ (*OSM*). This graph $G(E, V)$ models in its vertices (V) the intersections of streets and its edges (E) the city's streets.

To formalize this approximation, we take the Eq. 2. Where D_x represents the sum of the demand for $x = \{\text{surgeries, medical care, outpatient care, crush syndrome, others}\}$, for each district fraction d within the hospital's area of influence D_v for all district fractions N_v . It is essential to notice that the sum of each area fraction is taken in Km^2 by the factor calculated $fact_x$. This factor corresponds to the proportion of the population treated within 12, 24, 48 and 72 h after the earthquake.

$$D_x = \sum_{d \in D_v}^{N_v} (d \times fact_x) \quad (2)$$

Finally, the total demand for a hospital is given by the vector: $DT_v = [D_{Surgeries}, D_{Medicalcare}, D_{Outpatientcare}, D_{Crushsyndrome}, D_{Others}]$.

3) *Hospital accessibility estimation.*- For the calculation of the roads' accessibility, we take the subgraph $G_v(V, E) \subset G(V, E)$ corresponding to a hospital's area of influence. Based on $G_v(V, E)$, we compute the time it takes to get from an extreme point in the region of influence to the hospital using the Dijkstra [9] algorithm. It is important to note that the edges of the network used are weighted by the time it takes to drive in the street in question. This information was taken from Waze's SDK^{2,3}. For measuring access time to hospitals,

¹ OSM API: https://wiki.openstreetmap.org/wiki/API_v0.6.

² SDK Waze: <https://www.waze.com/es-419/sdk>.

³ Waze API <https://github.com/kovacsbalu/WazeRouteCalculator/blob/>.

road degradation was simulated by randomly removing up to 30% of the edges (*i.e.* streets), as formalized in the Eq. 3.

$$R_c = \frac{1}{|P|} \sum_{p \in P}^{|P|} (dijkstra(p, H)) \quad (3)$$

Where R is the average time with a $c\%$ degraded roads, p is a node in the P set of the outermost nodes, and H is the node where the hospital is in the area of influence.

Finally, when we measure the access time to hospitals and if it is indeed possible to reach them, we can estimate the number of people who die because they cannot reach the hospital in less than an hour. We focus on the second peak of trauma after instantaneous deaths, which occurs within minutes to several hours after the injury. Deaths in this period are usually due to epidural and subdural hematomas, hemopneumothorax, ruptured spleen, liver lacerations, pelvic fractures, or multiple other injuries associated with significant blood loss. **The golden hour** of post-injury care is characterized by the need for rapid assessment and resuscitation, which are the fundamental principles of advanced trauma life support [22]. It is important to emphasize that in the present study, we make the hypothesis that the distribution of the density of injured is uniform. The third peak, which occurs days after the initial injury, is often due to sepsis and multi-organ dysfunction. The care provided during each of the previous periods affects the results during this stage.

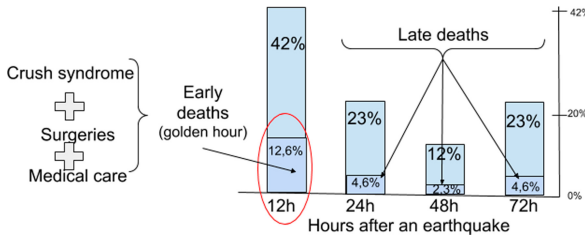


Fig. 1. Distribution of the percentages of need for hospital care according to the hours after the earthquake and the percentage of care within the golden hour.

Depending on the various traumas distribution, 30% correspond to those that must be treated in less than an hour [4, 21]. Thus, the Fig. 1 outlines the percentage of people with crushing, trauma, and need for surgical procedures of the total (*i.e.* 42%) of the total demand for hospital care within the first hours after the earthquake, need to be attended to in less than one hour (*i.e.* 12.6%). In this sense, the estimated deaths are proportional to the trips reaching the hospital despite road access degradation. As for the third peak, the amount of severe trauma represents 20% of the need for hospital care after 12h of the earthquake. This represents 4.6%, .3%, and 4.6% of the demand to be met after

24, 48 and 72 h respectively (*c.f.*, Fig. 1). In the following section, we describe the results obtained by applying the methodology described in this chapter.

4 Results

In this section, we describe the obtained results for different HTC hospital rankings and autonomy attention hours. It is worth noting that we take the HTC and the reinforced HTC, which considers the surgical procedures buildings plus the capacity for medical care and outpatient care, among others. Thus, we have the hospital care capacity scenario after 12 h. (*E12*), which considers only the supply of surgical procedures and another that also considers the supply of medical care, outpatient care, and others (*E12R*).

In the Table 2, we detail the index of each of the hospitals that would intervene in the different scenarios. We can see that depending on the HTC different hospitals are selected in each scenario. For example, if we take scenarios E12 and E12R with capacities *HTC12* and *HTC12R*, the hospitals in common are the ones that have indexes 1, 2, 3, 4, 5, 9, 10, 20, 21, 22, 23, 25, 35, 36, and 41. We also note that hospitals that are unique in *E12* are 8, 13, 24, 29, and 33; and the only ones in *E12R* are 15, 16, 19, 38, and 40.

Table 2. Hospital facilities code in different scenarios.

Scenario	Hospital facilities code
E12	22, 23, 35, 4, 5, 21, 36, 41, 9, 1, 20, 10, 24, 25, 2, 3, 29, 33, 8, 13
E12R	23, 9, 5, 4, 10, 21, 35, 1, 22, 2, 3, 20, 38, 41, 36, 40, 15, 25, 19, 16
E24	23, 22, 4, 35, 5, 21, 41, 36, 9, 1, 25, 10, 20, 3, 29, 7, 16, 40, 6, 19
E24R	23, 9, 5, 4, 10, 1, 35, 21, 22, 3, 20, 38, 41, 36, 40, 25, 19, 12, 16, 7
E48	23, 4, 35, 21, 36, 41, 1, 20, 10, 25, 3, 16, 7, 6, 12, 26, 38, 27
E48R	23, 4, 10, 1, 35, 21, 3, 20, 38, 41, 36, 25, 12, 16, 7, 14, 27, 26, 6, 11
E72	35, 41, 36, 1, 25, 20, 16, 26, 12, 38, 27
E72R	1, 35, 20, 38, 41, 36, 25, 12, 16, 27, 26

Now that the scenarios have been defined, we will calculate the demand for hospital care by each hospital in the established scenario. One of the problems with the works found in the literature is that they calculate the demand according to the political division *i.e.*, districts. However, after an earthquake, people do not necessarily think about going to their district hospital because it may be geographically farther away or because it may be inoperable after the earthquake. In that sense, we assume that people go to the nearest open hospital.

This way, we take the hospitals in each scenario, and from the 20 hospitals that offer the most *HTC*. So, we calculated the area of influence using Voronoi diagrams. The Figs. 2A, 2B, 2C, 2D, 2E, 2F, 2G, and 2H shows the area of influence of hospitals in scenarios E12, E12R, E24, E24R, E48, E48R, E72, and E72R,

respectively. These areas of influence intersected with the Lima graph allow us to find the areas of influence in Km^2 to find the demand for the number of surgeries, hospital care, outpatient care, people with crush syndrome, and other care. Note that the demand for hospital care also has minimum and maximum levels (*c.f.*, Chart 1). For this reason, we decided to additionally include an optimistic estimate of the demand for hospital care following an 8 Mw magnitude earthquake.

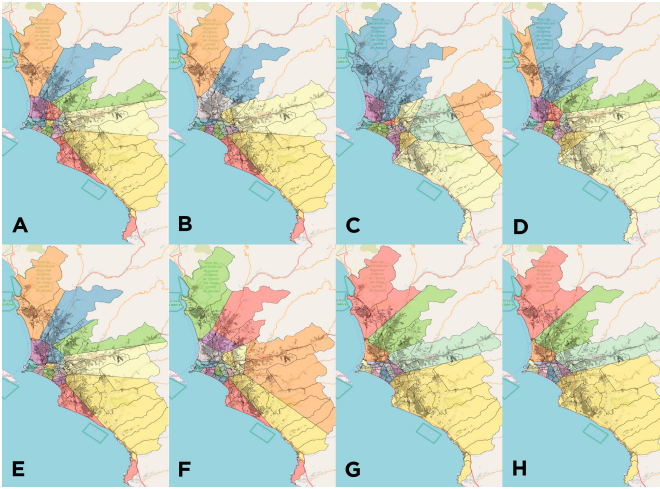


Fig. 2. Voronoi diagrams intersected with districts for the scenarios (A) E12, (B) E12R, (C) E24, (D) E24R, (E) E48, (F) E48R, (G) E72, and (H) E72R.

Table 3 summarizes the demands for hospital care measured in numbers of people for all scenarios and also shows the different capacities of hospital care in numbers of people that can be served. It should be noted that the demands for hospital care do not vary practically between the simple hospital care capacity scenarios (*E*) and reinforced (*ER*) when we compare the Operations columns against the hospital care capacity in Table 3. We observe that an average of 15% of the demand for operations is covered in the optimistic scenario (*i.e.* EO) at different times. In this sense, if we consider the total demand and contrast it with the reinforced hospital care capacity (*i.e.* HPCR), the coverage drops to 1% of the demand for care. This situation becomes critical when we take into account the pessimistic scenario (*i.e.* EP) for different hours. For such a scenario, the hedge falls to 0.63% and 0.32%. That is, less than 1% of the demand for hospital care is met.

As far as hospital accessibility analysis is concerned. First, we take the network corresponding to a region of influence. Over this region, we choose the most external nodes to the network so that, through the Dijkstra algorithm, we measure the time of arrival at the hospital from a certain point. Now for each graph,

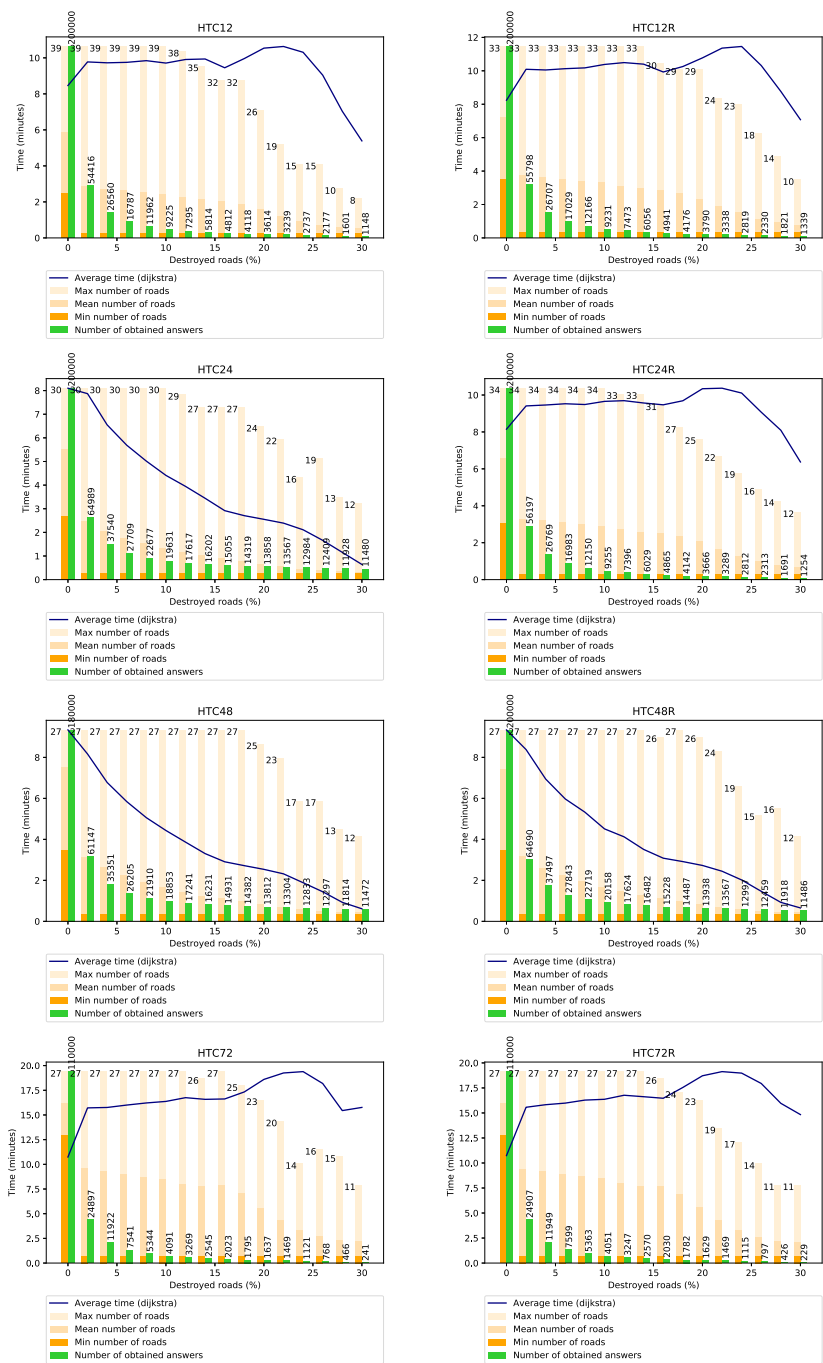


Fig. 3. Summary of the simulation of accessibility times for hospitals corresponding to all HTC scenarios.

Table 3. Summary of hospital care demands in number of people for the optimistic (OE) and pessimistic (PE) scenarios of 12, 24, 48 and 72 h after an 8 Mw magnitude earthquake. Where R, corresponds to the reinforced scenario (which takes into account infrastructure for outpatient care, hospitalization, crushing and others). The column ‘Total accumulated demand’ reports the need for accumulated care for the 12, 24, 48 and 72 h elapsed.

Scenario	Demand	Operations	Hospitable	Ambulatory	Flattening	Others	Total demand acum	HTC	HTCR
EO12	41%	1107	2079	8071	3755	23230	38242	150	318
EO24	65%	1714	3217	12489	5807	35946	59173	284	644
EO48	77%	2028	3811	14586	6881	42581	69887	424	1025
EO72	100%	2634	4949	18603	8933	55298	90417	321	798
EO12R	41%	1107	2079	8072	3753	23229	38240	143	346
EO24R	65%	1714	3218	12091	5804	35946	58773	281	672
EO48R	77%	2027	3811	14189	6876	42578	69481	424	1067
EO72R	100%	2633	4949	18206	8928	55295	90011	321	798
EP12	41%	27638	51969	24460	3752	23230	131049	150	318
EP24	65%	42778	80431	37851	5804	35946	202810	284	644
EP48	77%	50678	95282	44838	6876	42581	240255	424	1025
EP72	100%	65819	123743	58229	8928	55298	312017	321	798
EP12R	41%	27638	51969	24460	3752	23230	131049	143	346
EP24R	65%	42779	80430	37851	5804	35947	202811	281	672
EP48R	77%	50679	95281	44838	6876	42582	240256	424	1067
EP72R	100%	65820	123742	58229	8928	55299	312018	321	798

we degrade the roads by removing 2% to 30% from the streets. Thus for each level of degradation, 10 000 simulations were performed by the Monte Carlo method to measure the average time of arrival between the external points to the hospital. We measured the degradation level of the roads and the probability distribution of arrival times at the hospital for all scenarios from these simulations.

On the one hand, Fig. 3 shows a blue line indicating the average time - in minutes - it takes to reach the hospital from outside the hospital’s region of influence as a function of the percentage of roads destroyed. However, we see that the curve falls when the percentage of destruction is higher. This phenomenon can be explained by looking at the orange and green bars. Thus, the orange bars measure the number of available routes and the green ones the number of trips that reach the hospital.

On the other hand, concerning the distribution of the probability of arrival times at the hospital, for each of the trips, we make that return a positive value, i.e., a route is found, we show the distribution of the probability of arrival times at the hospital. As can be observed in Fig. 4, the distribution of arrival times shows a form of decreasing. Thus, we record the ratio of p of successful trips about the number of expected simulations, the mean, median, and standard deviation for each level of degradation. We can observe that the distribution shifts to the right as the destruction increases. However, from 16% of destruction, we see that

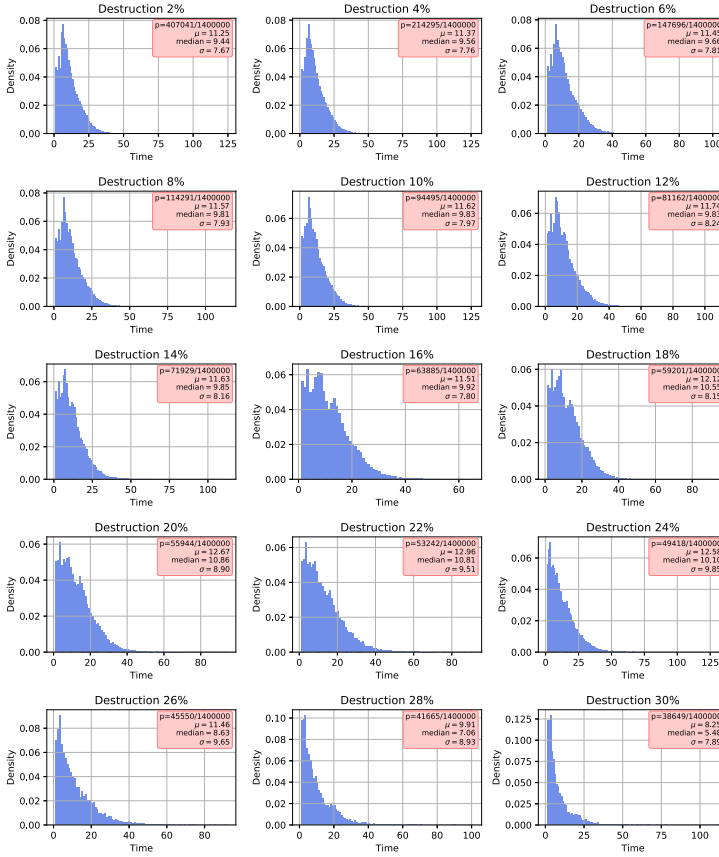


Fig. 4. Distribution of accessibility times to hospitals in different scenarios. (Color figure online)

the mean falls as the p ratio becomes smaller due to isolated nodes without being able to reach the hospital. Only the closest external nodes remain accessible.

Based on the results of accessibility times (*c.f.*, Fig. 3), it is possible to estimate the percentage of trips that do not reach the hospital due to the percentage of roads degradation. Thus, taking into account the total demand for hospital care composed of surgical interventions, hospital care, and crushing syndromes, we have a total 194 586 of people who needed care. Of this total, 42% will need attention in the first twelve hours, and of that total 30% is considered early death. That is, they need to be attended to in less than an hour. Thus, 24 518 people who needed attention in the first hour after the earthquake. Then, we calculated the percentage that does not arrive due to the road's degradation. Figure 5 illustrates the percentage of deaths according to the level of destruction of access to the hospital for different scenarios. The results range from 16 189 people passed away in the scenario with less destruction, which is 2% until 24 467

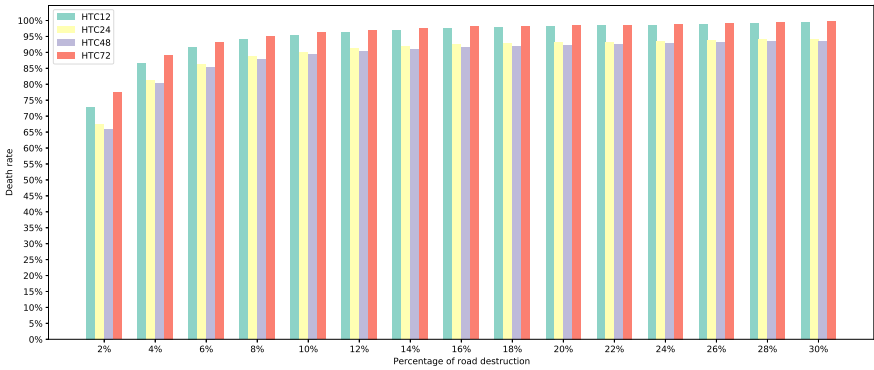


Fig. 5. Number of deaths caused by the inability to reach the hospital within the golden hour.

people passed away than when access degradation reaches 30% of the hospital's influence area. This must be added to the late deaths, that is, after the first 12 h of the earthquake.

In that sense, Table 4 summarizes the final demands after the considered deaths. It is essential to consider that we position ourselves in the worst scenario where 30% of the roads were destroyed. Thus practically 100% of people who need help both in the golden hour and after twelve hours in the late death phase

Table 4. Summary of hospital care demands expressed in the number of people for the optimistic (EO) and pessimistic (PE) scenarios for 12, 24, 48, and 72 h after an 8 MW earthquake. where *R* corresponds to a reinforced scenario (which takes into account infrastructure for outpatient care, hospitalizations, crushing, and others).

Scenario	Demand	Operations	Hospit-able	Ambula-tory	Flattening	Others	Total demand acum	Deaths	Deaths acum	Final demand	HTC	HTCR
EO12	41%	1107	2079	8071	3755	23230	38242	2082	2082	36160	150	318
EO24	65%	1714	3217	12489	5807	35946	59173	971	3053	56120	284	644
EO48	77%	2028	3811	14586	6881	42581	69887	249	3302	66585	424	1025
EO72	100%	2634	4949	18603	8933	55298	90417	953	4255	86162	321	798
EO12R	41%	1107	2079	8072	3753	23230	38240	2082	10014	28226	143	346
EO24R	65%	1714	3218	12091	5804	23514	58773	497	10511	48262	281	672
EO48R	77%	2027	3811	14189	6876	23938	69481	476	10987	58494	424	1067
EO72R	100%	2633	4949	18206	8928	24259	90011	1904	12892	77119	321	798
EP12	41%	27638	51969	24460	3752	23230	131049	25008	29898	101151	150	318
EP24	65%	42778	80431	37851	5804	35946	202810	3330	33227	169583	284	644
EP48	77%	50678	95282	44838	6876	42581	240255	869	34096	206159	424	1025
EP72	100%	65819	123743	58229	8928	55298	312017	3330	37426	274591	321	798
EP12R	41%	27638	51969	24460	3752	23230	131049	25008	29898	101151	143	346
EP24R	65%	42779	80430	37851	5804	35947	202811	3330	33227	169584	281	672
EP48R	77%	50679	95281	44838	6876	42582	240256	869	34096	206160	424	1067
EP72R	100%	65820	123742	58229	8928	55299	312018	3330	37426	274592	321	798

cannot reach the nearest hospital. The column *Demanda* shows the percentage of care that must be carried out. In this way, we cumulatively represent the need for care according to how time passes at 12, 24, 48, and 72 h after the earthquake with the different hospitals changing according to the optimistic scenario (OS) or pessimistic scenario (PS). The evolution of the scenario after the magnitude 8 Mw earthquake is alarming since, in the best of the EO48R scenarios and discounting deaths due to not reaching the hospitals, barely 2% of the demand for care is covered more catastrophic scenarios of 0.3% care coverage. These figures reflect the fragility of the Peruvian system.

5 Conclusions

This paper analyzes the hospital system in metropolitan Lima in the face of an 8 MW magnitude earthquake scenario to assess the gaps between the need for hospital care and the capacity of hospital care. For this analysis, we took 41 category II and III hospitals from the health network of MINSA and EsSalud, where we calculated the hospital care capacity (HTC). Thus, we took the 20 hospitals with the best care capacity for four different scenarios based on the number of hours after the earthquake. At this point, we found that hospitals significantly vary when considering the time of autonomy they have. This means that not all hospitals are prepared to provide care for 12, 24, 48, or 72 h autonomously after an earthquake.

Depending on the hospitals for different scenarios, we build the hospitals' influence zone to determine the number of people who need care. We then measure whether the hospital's care capacity (HTC) meets the minimum and maximum demand for hospital care. The results corroborate what we found in the literature, where the fragility of the health system in metropolitan Lima is evident. Thus, we simulated the hospitals' accessibility according to road degradation using the hospital's areas of influence. By simulating the degradation of the accessibility of the outermost points of the zones of influence to the hospital, we observe that several points in the zones of influence remain isolated and cannot reach the hospital because of the destroyed roads. This allowed us to observe that only the closest external points can access the hospital, with the degradation of 16%.

Finally, road degradation allowed us to quantify the number of deaths within the first twelve hours (early deaths) and after 24 h (late deaths) due to road degradation. Thus, we found that for degradation of 30% almost 100% of people do not reach the hospital and if they still do, the capacity of hospital care does not cover the large number of people who would reach the hospitals, since hospitals cover between 1% and 15% of the demand for hospital care depending on the scenario.

It would be interesting to integrate population density dynamics in the hospitals' demand estimation using Call Detail Records data for changes as a function of day and time due to population mobility. Also, modeling the queues for the same care services would provide a more accurate picture of actual care capacity.

Acknowledgment. We would like to thank our colleagues Gabriel Aguirre Martens and Hugo Alatrística-Salas for the discussion, review and suggestions that enrich the present work.

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